Venkata Lakshmi Parimala Pasupuleti

Software Engineer · LLM Evaluation Infrastructure · Python · AgentProbe · CI/CD · Observability

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Software Engineer — Siri Evaluation Infrastructure · Apple

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Dear Hiring Team,

I built AgentProbe because I needed it. While developing LeaseI — an 8-agent AI platform for property operations — I had no reliable way to measure whether the system was actually getting better across iterations. So I built the infrastructure: a CI-integrated 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-judge scoring, golden dataset regression harness, and distributed model routing across OpenAI/Anthropic/Groq/Ollama. That infrastructure is now open source. It is also, as far as I can tell, exactly what this role is tasked with building for Siri.

The design philosophy I applied to both AgentProbe and LeaseI maps directly to what you describe as “evals first.” For LeaseI, I didn’t declare success by running the system on a demo — I built a pass@k harness with multi-trial sampling, LLM-as-judge scoring, and a regression gate before shipping anything: 93.6% precision, 87.9% F1, pass@1 81.7% / pass@3 88.3% across 1,240 lease PDFs covering 5 US states and federal regulations. The core design rule I enforced in code: LLM extracts and reasons; deterministic logic or a human makes every final regulated decision. That boundary is measurable, testable, and reproducible — which is the only kind of AI quality signal worth trusting.

The backend depth is there too. At Entrata, I spent six years instrumenting production systems with OpenTelemetry, Prometheus, and Grafana for 100K+ monthly active users at 99.5% uptime — building operational dashboards, alert logic, and debugging tooling that cut mean time-to-diagnose by 40% and kept the bug count below 10 for six consecutive months. I owned failures through to permanent root-cause fixes, not workarounds. At GMU, I fine-tuned a custom classification model using HuggingFace Transformers and PyTorch, ran the full eval pipeline, and presented training methodology and results to faculty — the same eval-science rigor the preferred qualifications describe.

Siri is one of the most widely deployed AI systems in the world. Building the infrastructure that keeps its quality signal trustworthy — before features ship, not after — is exactly the kind of work I want to do. I’d welcome the chance to talk through how AgentProbe and LeaseI were built and what that means for this role.

Sincerely,

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